Nuclear Astronomy and Project Hail Mary

The Petrova Line problem

In the 2026 film *Project Hail Mary* (originally based a science-fiction novel written by Andy Weir) Explore terrifying astronomical mystery to humans. The Sun is slowly losing its luminosity (dimming) The culprit is microorganism like extraterrestrial lifeform called **Astrophage** that thrives in the solar atmosphere and consumes solar energy at a rate that threatens the global climate. These organisms spread through space and feed on stellar radiation, effectively reducing the energy output of stars over time. Since Earths climate and biosphere depend entirely on solar energy even a slight decrease in solar luminosity would trigger global cooling and eventually the extinction of human civilization. The mystery begins when astronomer Irina Petrova identifies a strange infrared arc extending from the Sun toward Venus as shown in Figure 1. It becomes the first major clue that something unnatural is happening to the Sun and nearby stars and it is called Petrova Line (by the name of the founding astronomer). In the world of astronomical spectroscopy, stellar light typically follows the predictable laws of blackbody radiation and atomic absorption. However, the Petrova Line represents an anomalous infrared absorption feature where Astrophage are actively harvesting energy.

Figure 1: The Petrova Line is a mysterious spectral signature discovered by the scientist Irina Petrova

The global response to save human race was the *Project Hail Mary* which is an interstellar mission to Tau Ceti, the only nearby star seemingly resistant to this infection. The mission's earth's main charachter is Ryland Grace, a middle-school teacher and PhD molecular biologist. His unique expertise is required because the threat is not merely an astronomical phenomenon but a biological one. Humanity needed someone who could understand Astrophage, an alien microorganism capable of absorbing and storing enormous amounts of energy. Since Astrophage behaved like a living organism rather than a machine or physical phenomenon, expertise in biology became essential. As a result, Graces combined background in biology engineering intuition, and teaching ability makes him uniquely suited for the mission. This represents a strong example of interdisciplinary science (Like nuclear astrophysics) where biology, astrophysics, and engineering converge to save humankind.

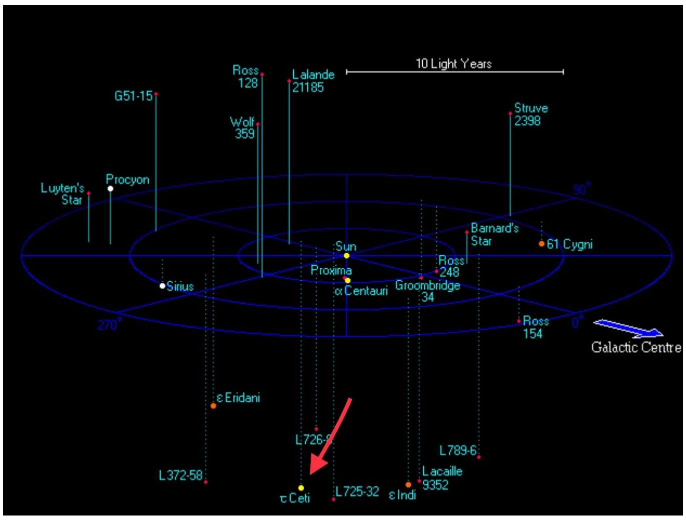

Figure 2: Distance between the Solar System and Tau Ceti ($\sim$ 11.9 light years). For Ryland Grace, this represents a one way interstellar journey

I do not wish to spoil this fascinating and imaginative story anymore. Instead, I encourage all of you explore both the book and the film version of *Project Hail Mary.* However, in this assignment we focus on the Conceptual scientific problem. We will apply concepts from nuclear astrophysics and stellar structure to investigate the physical implications of the Petrova Line problem.

The Virial Response (Derivation & Theory)

Energy Drain

The Astrophage migration creates a massive energy sink on the Suns surface. Let the Sun's total energy be $E_{tot} = E_{KE} + E_{GR}$. If Astrophage consume energy faster than the core produces it via fusion, the star radiates (or loses) a net amount of energy ΔE. Using the non-relativistic Virial Theorem ($2E_{KE}+E_{GR} = 0$), derive an expression for the change in internal kinetic energy (ΔE_{KE}) in terms of the total energy change (ΔE_{tot})?

Thermostat Effect

Based on your derivation, explain why the Sun's core temperature T (where $E_{KE} \propto T$) must *increase* as a result of the Astrophage stealing its energy. Why is this behavior described as having a negative heat capacity?

Contraction

To satisfy the Virial Theorem, the Sun must move to a more "bound" state. Prove mathematically that if E_{KE} increases, the radius R must decrease (Recall $E_{GR} \approx -GM^2/R$)

The Habitable Zone Paradox

If the Astrophage drain energy from the surface, the *observed* luminosity L reaching Earth drops, causing an ice age in Earth. Based on your above proofs explain how this increased core temperature eventually leads to higher fusion rates. Is this enough to stop the collapse? In the novel, the Sun stays yellow but dims. However, if the core continues to heat and contract, we approach the stability limit defined by the adiabatic index $\gamma = 4/3$. Explain why transitioning to an ultra-relativistic state ($\gamma \to 4/3$) would make the Sun's equilibrium "marginally stable" and potentially lead to a catastrophic expansion or collapse.

Critical Analysis

Assume our Sun emits 10^{26} Joules per second. If Astrophage consume 0.01% of the Sun's total luminosity, calculate the energy lost in one Earth year. Using the free-fall time-scale $t_{ff} = \sqrt{3\pi/32G\rho}$, calculate how fast the Sun would collapse if the Astrophage theoretically consumed *all* internal pressure support ($P \to 0$) instantly. (Use $\rho_{\odot} \approx 1400$ kg/m^3).